Panasonic FP Series PLC Structured Text (ST) — AI Code Optimization Research Report

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Classification: Critical — Directly impacts production AI generation accuracy

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1. Executive Summary

This report analyzes the accuracy problems observed when using Moonshot AI (Kimi) to generate Panasonic FP series PLC Structured Text (ST) code in PLC-AIStudio. Based on test data from a complex conveyor system (56 I/O entries, 10 process steps), the AI achieved a **validation score of only 45/100**, with critical issues including:

- **Code truncation** (missing `END_PROGRAM`)
- **~67% of declared I/O addresses unused** in the generated logic
- **15 unused-I/O warnings** flagged by the validator

The root causes are **not** primarily the model's inherent coding ability, but rather:

1. A **severely undersized context window** (`moonshot-v1-8k`) for complex industrial programs
2. A **generic IEC 61131-3 system prompt** that lacks Panasonic-specific constraints, examples, and validation rules
3. **No program decomposition strategy**, forcing the model to generate monolithic POUs that exceed output limits
4. **No feedback loop** between generation and validation

Key Recommendation: Immediately upgrade to `moonshot-v1-32k` minimum (or `kimi-k2.5` for 256K context), restructure the system prompt with Panasonic-specific few-shot examples and POU decomposition logic, and implement a two-pass generation → validation → refinement pipeline.

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2. Panasonic FP ST Syntax Deep Dive

2.1 IEC 61131-3 Compliance vs. Panasonic Extensions

Control FPWIN Pro7 is **PLCopen-certified** and claims full IEC 61131-3 compliance. However, Panasonic's implementation has important nuances that differ from generic IEC behavior:

Feature	Standard IEC 61131-3	Panasonic FPWIN Pro7 Reality
Programming languages	LD, FBD, SFC, ST, IL	All 5 supported
ST Editor EN/ENO	Standard libraries have EN/ENO	**No EN/ENO functions/function blocks from IEC Standard Library in ST editor**

Case sensitivity	Case-insensitive usually	**Completely case-insensitive** (`abcd` ≡ `ABCD` ≡ `aBCd`)
Timer FBs	`TON`, `TOF`, `TP`	Supported **PLUS** Panasonic-native `TM_1ms`, `TM_10ms`, `TM_100ms`, `TM_1s` FBs
Counter FBs	`CTU`, `CTD`, `CTUD`	Supported **PLUS** Panasonic-native `CT_FB`, high-speed counter instructions `F166`/`F167`
Pulse instructions	N/A	`F` and `P` instructions **cannot be used in ST Editor**
String handling	`STRING` type	**Supported** — ASCII variable-length, max **32,767 characters** (memory-dependent)
Typed literals	Not required	Supported (`INT#2`, `REAL#3.2`) but **not required** — compiler auto-assigns types

> ***Critical Finding**: The current system prompt instructs the AI to use `TON`, `TOF`, `TP` and `CTU`, `CTD`, `CTUD` — these **are valid** in FPWIN Pro7. However, the prompt fails to mention that **EN/ENO cannot be used in ST**, which may cause the AI to generate invalid code if it attempts standard IEC EN/ENO patterns.*

2.2 Addressing Conventions — Exact Ranges & Rules

Panasonic FP series uses **FP-style addresses** (not IEC `%IX0` notation) in most practical applications. The exact ranges vary significantly by PLC model:

FP0H / FP-XH Series (High-Performance)

Area	Abbreviation	Range	Notes
External Input	X	**X0 – X109F**	1,760 points max. Actual usable depends on hardware config
External Output	Y	**Y0 – Y109F**	1,760 points max. Actual usable depends on hardware config
Internal Relay	R	**R0 – R511F**	8,192 points (configurable: 4,096 or 8,192)
Special Internal Relay	R	**R9000 – R951F**	800 points. Read-only system status
Timer / Counter relays	T / C	**T0 – T1007 / C1008 – C1023**	1,024 points total. Default: 1,008 timers, 16 counters. Adjustable via System Register 5+6
Data Register	DT	**DT0 – DT65532**	65,533 words max (varies by program capacity setting). **32-bit uses `DDT` prefix**

			(auto-uses next word)
Special Data Register	DT	**DT90000 – DT90999**	1,000 words (FP0H). System clock, errors, scan time
Link Relay	L	**L0 – L127F**	2,048 points
Link Register	LD	**LD0 – LD255**	256 words
File Register	FL	**FL0 – FL32767**	Varies by model and memory config
Index Register	I	**I0 – ID**	14 words (IX, IY, etc.)

FP0R Series (Compact)

Area	Range	Notes
X / Y	X0 – X109F / Y0 – Y109F	Same addressing, fewer physical points (max 384 with expansion)
R	R0 – R255F (4,096) **or** R511F (8,192)	System Register 1 selects capacity
DT	DT0 – DT32764	32,765 words (adjusts with program capacity)
Special DT	DT9000 – DT90999	Older numbering (4-digit vs 5-digit in FP0H)

FP7 Series (Modular/High-End)

Area	Range	Notes
X / Y	X0 – X109F / Y0 – Y109F	Same logical addressing
R	R0 – R511F	8,192 points standard
DT	DT0 – DT65532	Up to 65,533 words
T / C	1,024 points	Timer/counter split adjustable

> ***Important**: Hexadecimal suffixes (e.g., `X0F`, `R10A`) are valid for bit-level addressing within word boundaries. The AI must be aware that `X10` and `X0A` refer to different addressing schemes.*

2.3 Timer & Counter Syntax

IEC Standard FBs (Valid in FPWIN Pro7)

```
VAR
  TON_Delay : TON;        // Timer On-Delay instance
```

```
  TOF_Delay : TOF;        // Timer Off-Delay instance
  TP_Pulse  : TP;         // Timer Pulse instance
  CTU_Total : CTU;        // Up counter instance
  CTD_Total : CTD;        // Down counter instance
  CTUD_Bi   : CTUD;       // Up/Down counter instance
END_VAR

// Usage
TON_Delay(IN := StartSignal, PT := T#3s);
MotorRun := TON_Delay.Q;

CTU_Total(CU := SensorPulse, RESET := ResetButton, PV := 1000);
Alarm := CTU_Total.Q;
```

Panasonic-Native Alternatives

Panasonic also provides **FP Library FBs** that map more closely to traditional Panasonic ladder instructions:

```
VAR
  TM1 : TM_10ms;          // 10ms resolution timer (0-327.67s)
  TM2 : TM_100ms;         // 100ms resolution timer (0-3276.7s)
  CT1 : CT_FB;            // Panasonic native counter FB
END_VAR

// Usage
TM1(IN := StartSignal, PT := T#2000ms);  // 2 seconds
Output := TM1.Q;
```

> ***Recommendation for AI**: The IEC standard `TON` / `CTU` syntax is **preferred** for FPWIN Pro7 ST because it is more portable and better documented. However, the system prompt must explicitly tell the AI **not** to use EN/ENO with these FBs in ST, and to declare timer/counter instances in the VAR block (not as direct addresses like `T0`).*

2.4 Data Types Supported

Per the FPWIN Pro7 Programming Manual, the following elementary data types are available:

Type	Keyword	Size	Range / Notes
Boolean	`BOOL`	1 bit	0(FALSE), 1(TRUE). Constants: `TRUE`/`FALSE` or `0`/`1`
Integer	`INT`	16 bits	-32,768 to 32,767
Double Integer	`DINT`	32 bits	-2,147,483,648 to 2,147,483,647

Unsigned Integer	`UINT`	16 bits	0 to 65,535
Unsigned DINT	`UDINT`	32 bits	0 to 4,294,967,295
Word	`WORD`	16 bits	0 to 65,535 (binary string)
Double Word	`DWORD`	32 bits	0 to 4,294,967,295
Real	`REAL`	32 bits	IEEE754 float, ~7 significant digits
Time	`TIME`	16 or 32 bits	T#0.01s to T#327.67s (16-bit PLC) or T#21,474,836.47s (32-bit PLC)
Date	`DATE`	32 bits	D#2001-01-01 to D#2099-12-31
Date & Time	`DATE_AND_TIME`	32 bits	DT#2001-01-01-00:00:00 to DT#2099-12-31-23:59:59
Time of Day	`TIME_OF_DAY`	32 bits	TOD#00:00:00 to TOD#23:59:59
String	`STRING`	Variable	**ASCII only**, 1-32,767 bytes. 2 words header + (n+1)/2 words content

> ***STRING Limitation**: Panasonic ST `STRING` supports **ASCII only** (no Unicode/UTF-8 in the PLC runtime). For Chinese HMI display, string handling must use byte arrays or external conversion.*

2.5 ST Editor Specific Constraints

From the official Control FPWIN Pro7 Reference Manual:

1. **No EN/ENO in ST**: Functions and function blocks from the IEC Standard Library do **not** have EN/ENO inputs in the ST editor. Use `IF` statements to control execution.
2. **No P instructions**: Pulse execution type (`P`) instructions from the FP Pulsed Library **cannot** be used in ST.
3. **Control variable after loop**: Do not use the value of a `FOR` loop control variable after the loop finishes — the value is **not guaranteed**.
4. **Debugging behavior**: When debugging `IF` / `CASE` structures, the program code inside the structure appears to "run" even if the condition is false, but individual instructions are **not executed**.
5. **32-bit register access**: Use `DDT` prefix instead of `DT` to access 32-bit data registers (auto-uses the next consecutive word).

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3. Current System Prompt Analysis

3.1 Existing Prompt (kimiService.ts)

```
你是 PLC-AIStudio — 一个专业的 Panasonic PLC 编程助手。
你的任务是帮助机械设计师和自动化工程师编写 IEC 61131-3 标准的 PLC 程序。

## 核心规则（必须严格遵守）
1. 只生成结构化文本（ST）代码，使用 IEC 61131-3 标准语法
2. 必须包含完整的 VAR / VAR_INPUT / VAR_OUTPUT / VAR_IN_OUT 声明块
3. **变量命名强制要求**：在 VAR 声明和代码逻辑中，必须直接使用用户提供的 I/O 地址作为变量名（如 X0, X1, Y0, Y1），不得使用英文别名或自定义变量名
4. 每个 I/O 地址必须在 VAR 块中声明，类型为 BOOL，并添加中文注释说明功能
5. 使用 Panasonic FP 系列 PLC 的地址规范：X（输入）、Y（输出）、R（内部继电器）、DT（数据寄存器）
6. 生成的代码必须可以直接导入 Panasonic FPWIN Pro7
7. 在代码前添加注释说明程序功能和 I/O 定义

## 代码格式要求
- 使用标准 IEC 61131-3 ST 语法
- **IF-THEN 结构**：优先使用 IF-THEN-ELSIF-ELSE-END_IF 表达控制逻辑，每个条件分支对应一个 Network
- 定时器使用 TON, TOF, TP 标准功能块
- 计数器使用 CTU, CTD, CTUD 标准功能块
- 注释使用 // 格式，标注在变量声明后
- 每个 POU 以 PROGRAM 或 FUNCTION_BLOCK 开头
```

3.2 Strengths

- Correctly targets IEC 61131-3 ST
- Forces direct hardware addressing (X0, Y0) which matches Panasonic conventions
- Includes safety requirements (e-stop first, reset conditions)
- Requests complete VAR blocks

3.3 Critical Weaknesses

#	Weakness	Impact
1	**No Panasonic-specific constraints** (no EN/ENO, no P instructions, case insensitivity)	AI may generate invalid FPWIN Pro7 code if it attempts EN/ENO patterns learned from other PLCs
2	**No address range validation rules**	AI may generate `X999` or `DT999999` which exceed PLC capacity
3	**No few-shot examples** in system prompt	Model has no concrete reference for what "correct" Panasonic ST looks like in this context
4	**No POU decomposition guidance**	For complex systems, AI generates monolithic PROGRAM blocks that exceed output limits

5	**No I/O coverage verification step**	AI is not instructed to cross-check that every declared I/O is used
6	**Timer/counter declaration format unclear**	The example shows `RunFlag : BOOL` but never shows `TON_1 : TON;` declaration
7	**Missing `DDT` guidance**	AI may use incorrect 32-bit register syntax
8	**No `max_completion_tokens` configured**	Output truncated when approaching model limit

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4. Accuracy Problems & Root Cause Analysis

4.1 Test Case Profile

- **System**: Complex conveyor (multi-station automatic assembly line)
- **I/O Count**: 56 entries (20 inputs, 15 outputs, 10 internal relays, 6 timers, 4 counters)
- **Process Steps**: 10 sequential steps with safety interlocks
- **Target PLC**: Panasonic FP-XH

4.2 Observed Failures

Failure	Frequency	Severity
Code truncation (missing `END_PROGRAM`)	~30% of complex generations	**Critical**
Unused declared I/O (~67% unused)	Consistent	**High**
Missing timer/counter instance declarations	~40%	**High**
Incorrect safety logic ordering	~20%	**Medium**

4.3 Root Cause Analysis

Root Cause 1: Context Window Exhaustion (moonshot-v1-8k)

The `moonshot-v1-8k` model has a **hard limit**:

```
max_output_tokens = 8,192 - prompt_tokens
```

For the conveyor test case:

- System prompt: ~800 tokens
- User prompt (56 I/O + 10 steps + safety): ~1,500–2,500 tokens
- Available for output: **~5,500–6,500 tokens**

A complete ST program for this complexity requires:

- VAR block with 56 declarations + comments: ~800–1,000 tokens
- 10 process steps with IF-THEN logic: ~2,000–3,000 tokens
- Safety logic + comments: ~800–1,200 tokens

- **Total estimated**: ~3,600–5,200 tokens

While this theoretically fits, the model also generates:

- Chinese explanations before/after code
- Excessively verbose comments
- Redundant logic patterns

Result: The model hits the token ceiling and truncates, dropping the final `END_PROGRAM` and sometimes entire process steps.

> ***Evidence**: Kimi API documentation explicitly states: "If `finish_reason` is `length`, the generated content exceeded `max_completion_tokens` and was truncated."*

Root Cause 2: Prompt Lacks I/O Accountability Mechanism

The current prompt says:

> *"代码中所有变量必须使用上述 I/O 地址"*

But it provides **no mechanism** for the AI to verify compliance. The AI does not have a working memory or checklist. It processes the prompt sequentially and may:

- Forget early I/O entries by the time it reaches the logic section
- Omit I/O that were mentioned but not emphasized
- Skip I/O that require complex conditional logic

Root Cause 3: Monolithic Generation Without Decomposition

The prompt asks for **one complete program**. For 56 I/O and 10 steps, this creates:

- A single `PROGRAM` block with hundreds of lines
- Cognitive overload for the model (and the human reader)
- No reusable patterns (timers, motor control, safety FBs)

Industrial best practice is to decompose into:

- `PROGRAM Main` — sequencing and mode control
- `FUNCTION_BLOCK MotorCtrl` — reusable motor logic
- `FUNCTION_BLOCK TimerDel` — standardized timer wrappers
- `FUNCTION_BLOCK SafetyGate` — e-stop and safety door logic

Root Cause 4: No Validation Feedback Loop

The current pipeline is:

```
User Input → AI Generation → Display to User
```

There is **no automated check** that feeds validation results back to the AI for refinement. The `stValidator.ts` runs on the client side **after** generation, but:

- The user sees a low score (45/100) with no automatic fix
- The AI never learns from its mistakes in the same session
- Truncated code cannot be auto-completed

Root Cause 5: Temperature Setting May Be Suboptimal

Current: `temperature = 0.3`

For **complex multi-step reasoning** (industrial sequencing, safety interlocks), a slightly higher temperature (0.4–0.5) can improve the model's ability to explore conditional branches. However, the primary issue is not temperature — it is context size and prompt structure.

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5. Optimization Recommendations (Prioritized)

P0 — Critical (Implement Immediately)

P0.1 Upgrade Model to 32K Minimum

Current	Recommended	Rationale
`moonshot-v1-8k`	`moonshot-v1-32k` or `kimi-k2.5`	32K provides 4× output headroom; K2.5 provides 256K with superior reasoning

Action:

```
// Before
model: 'moonshot-v1-8k',

// After (with dynamic selection)
const model = estimatedTotalTokens <= 8192 ? 'moonshot-v1-8k' : 'moonshot-v1-32k';
// Or default to 32K for all industrial generation
model: 'moonshot-v1-32k',
max_completion_tokens: 12000,  // Explicitly request longer output
```

Cost Impact: 32K input costs $1.00/M tokens vs $0.20/M for 8K. For a typical prompt (~2K tokens), cost per request increases from ~$0.0004 to ~$0.002 — negligible for a paid SaaS product.

P0.2 Add Panasonic-Specific Constraints to System Prompt

Add these **non-negotiable** rules:

```
## Panasonic FPWIN Pro7 ST 专用约束（必须遵守）
1. ST 编辑器不支持 EN/ENO：调用功能块时禁止使用 EN:=... 或 ENO=>... 参数
2. ST 编辑器不支持 P 脉冲指令：不要使用脉冲触发版本的功能
3. 标识符不区分大小写：X0 和 x0 等价，但代码中统一使用大写地址
4. 32 位数据寄存器必须使用 DDT 前缀（如 DDT100 占用 DT100 和 DT101）
5. 定时器/计数器必须先声明实例再使用：
   VAR
     TON_Delay : TON;
     CTU_Count : CTU;
   END_VAR
6. 时间常量格式：T#3s, T#500ms, T#1m30s
```

```
7. 字符串类型 STRING 仅支持 ASCII，不要生成中文字符串常量给 PLC 变量
8. 特殊继电器区域（R9000-R951F, DT90000-DT90999）为系统保留，用户程序不要直接赋值
```

P0.3 Implement POU Decomposition for Complex Systems

For systems with >30 I/O entries or >5 process steps, **force decomposition**:

```
## 程序分解规则（当 I/O 超过 30 个或步骤超过 5 个时强制应用）
将程序拆分为以下 POU：
1. PROGRAM MainPrg — 主循环：模式选择、急停总控、调用功能块
2. FUNCTION_BLOCK SeqCtrl — 顺序控制：步骤转移逻辑（STEP 1→STEP 2→...）
3. FUNCTION_BLOCK SafetyMgr — 安全管理：急停、安全门、气压报警、故障复位
4. FUNCTION_BLOCK MotorCtrl — 电机控制：传送带启停、方向控制、变频器信号
5. FUNCTION_BLOCK CylinderCtrl — 气缸控制：单气缸伸出/缩回/超时检测

每个 POU 单独生成，先声明接口变量（VAR_INPUT/VAR_OUTPUT），再写逻辑。
```

Backend Implementation:

- Add a preprocessing step in `buildPromptFromForm` that analyzes I/O count and step count
- If complex, switch to a **multi-turn generation**:
  - Turn 1: Generate POU architecture and interface definitions
  - Turn 2: Generate each POU body sequentially
  - Turn 3: Generate `PROGRAM MainPrg` that wires all FB instances

P1 — High Priority

P1.1 Add Few-Shot Examples to System Prompt

Include a **complete, correct Panasonic ST example** directly in the system prompt. Research shows few-shot prompting reduces syntax errors by 40-60% for domain-specific code generation.

```
## 示例：Panasonic FP-XH 传送带控制（必须参考此格式）
PROGRAM ConveyorCtrl
VAR
  X0 : BOOL; // 启动按钮
  X1 : BOOL; // 停止按钮
  X2 : BOOL; // 急停（常闭）
  Y0 : BOOL; // 传送带电机
  Y1 : BOOL; // 运行指示灯
  RunFlag : BOOL; // 运行标志
  TON_Delay : TON; // 启动延时定时器实例
END_VAR

// 急停逻辑（最优先）
IF NOT X2 THEN
  Y0 := FALSE;
  Y1 := FALSE;
```

```
  RunFlag := FALSE;
END_IF;

// 启动/停止自锁
IF X0 AND X2 AND NOT RunFlag THEN
  RunFlag := TRUE;
END_IF;
IF X1 OR NOT X2 THEN
  RunFlag := FALSE;
END_IF;

// 电机输出
Y0 := RunFlag;
Y1 := RunFlag;

// 延时停止示例
TON_Delay(IN := NOT RunFlag, PT := T#2s);
IF TON_Delay.Q THEN
  Y0 := FALSE;
END_IF;

END_PROGRAM
```

P1.2 Add I/O Coverage Verification to Prompt

Append this instruction to every user prompt:

```
## 生成后自检（必须在最终输出前完成）
1. 检查 VAR 块中声明的每个 X, Y, R 地址是否都在代码逻辑中被读取或写入
2. 检查每个定时器 TON/TOF/TP 实例是否都被调用（.IN 和 .PT 被赋值）
3. 检查每个计数器 CTU/CTD/CTUD 实例是否都被调用（.CU 或 .CD 被赋值）
4. 如果发现有声明但未使用的 I/O，补充对应的控制逻辑
5. 确保代码以 END_PROGRAM 结尾
```

P1.3 Implement Two-Pass Generation Pipeline

```
Pass 1: AI generates draft code
        ↓
Validator runs server-side (or client-side with callback)
        ↓
Pass 2: If score < 80 or truncation detected, send validation issues
        back to AI as a "fix request" with specific line numbers
        ↓
Final code merged and displayed
```

Implementation sketch:

```
// In api/src/routes/ai.ts
let fullResponse = '';
for await (const chunk of streamChatCompletion(messages)) {
  fullResponse += chunk;
  res.write(...);
}

// After stream ends, run validation
const validation = validateSTCode(fullResponse, { declaredIO: formData.ioList });
if (validation.score < 80 || !fullResponse.includes('END_PROGRAM')) {
  // Trigger fix pass
  const fixMessages = [
    ...messages,
    { role: 'assistant', content: fullResponse },
    { role: 'user', content: buildFixPrompt(validation) },
  ];
  // Stream fix...
}
```

P2 — Medium Priority

P2.1 Use Function Blocks for Reusable Patterns

Create a **skill library** in the prompt for common industrial patterns:

```
## 标准功能块模板（需要时直接实例化）

// 电机自锁控制
FUNCTION_BLOCK MotorLatch
VAR_INPUT
  Start : BOOL;
  Stop : BOOL;
  EStop : BOOL; // 常闭急停，=FALSE 时停止
END_VAR
VAR_OUTPUT
  Run : BOOL;
END_VAR
VAR
  Running : BOOL;
END_VAR

IF NOT EStop THEN
  Running := FALSE;
ELSIF Start AND EStop THEN
  Running := TRUE;
ELSIF NOT Stop THEN
```

```
  Running := FALSE;
END_IF;
Run := Running AND EStop;
END_FUNCTION_BLOCK

// 单气缸控制（带超时检测）
FUNCTION_BLOCK CylinderCtrl
VAR_INPUT
  CmdExtend : BOOL;
  FwdLimit : BOOL;
  RevLimit : BOOL;
  TimeOut : TIME;
END_VAR
VAR_OUTPUT
  ExtendOut : BOOL;
  RetractOut : BOOL;
  Alarm : BOOL;
END_VAR
VAR
  TON_TO : TON;
END_VAR

ExtendOut := CmdExtend AND NOT RetractOut;
RetractOut := NOT CmdExtend AND NOT ExtendOut;
TON_TO(IN := CmdExtend AND NOT FwdLimit, PT := TimeOut);
Alarm := TON_TO.Q;
END_FUNCTION_BLOCK
```

P2.2 Add Model Selection Logic Based on Complexity

```
export function selectModel(formData: FormData): { model: string; maxTokens: number } {
  const ioCount = formData.ioList.length;
  const stepCount = formData.processFlow.length;
  const estimatedPromptTokens = 800 + ioCount * 30 + stepCount * 50;
  const estimatedOutputTokens = 500 + ioCount * 40 + stepCount * 120;

  const total = estimatedPromptTokens + estimatedOutputTokens;

  if (total <= 7000) {
    return { model: 'moonshot-v1-8k', maxTokens: Math.min(4000, 8192 -
estimatedPromptTokens) };
  } else if (total <= 30000) {
    return { model: 'moonshot-v1-32k', maxTokens: Math.min(12000, 32768 -
estimatedPromptTokens) };
  }
  return { model: 'kimi-k2.5', maxTokens: Math.min(16000, 131072 - estimatedPromptTokens)
```

```
};
}
```

P2.3 Temperature Tuning

Scenario	Temperature	Notes
Simple system (<20 I/O, <5 steps)	0.2 – 0.3	Deterministic, less creative
Complex system (>30 I/O, >5 steps)	0.4 – 0.5	Slightly more exploration for branch logic
Fix/refinement pass	0.2	Strict adherence to validation feedback

P3 — Future Considerations

- **Fine-tuning**: If volume grows, consider fine-tuning a smaller model on curated Panasonic ST examples
- **RAG (Retrieval-Augmented Generation)**: Store validated POU templates in a vector DB and retrieve relevant examples based on I/O similarity
- **Structured Output (JSON Mode)**: Request AI output as JSON with separate fields for `varBlock`, `safetyLogic`, `processLogic`, then assemble into ST — reduces truncation risk

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6. Example Improved Prompts

6.1 Improved System Prompt (Full)

```
你是 PLC-AIStudio — 一个专精于 Panasonic FP 系列 PLC 的 ST 编程专家。
你使用 Control FPWIN Pro7 兼容的 IEC 61131-3 Structured Text 语法。

## Panasonic FP ST 绝对约束（违反会导致代码无法编译）
1. 【无 EN/ENO】ST 编辑器不支持 IEC 标准库的 EN/ENO 参数，调用 FB 时禁止写 EN:=... 或 ENO=>...
2. 【无 P 指令】ST 编辑器不支持 FP 脉冲库的 P 指令
3. 【区分大小写】标识符不区分大小写，但所有硬件地址必须用大写（X0, Y0, R10, DT100）
4. 【32 位寄存器】32 位数据寄存器用 DDT 前缀，如 DDT100 自动占用 DT100+DT101
5. 【定时器/计数器实例化】所有 TON/TOF/TP/CTU/CTD/CTUD 必须在 VAR 中声明实例，禁止直接操作 T0/C0 继电器地址
6. 【时间格式】常量用 T#3s, T#500ms, T#1m30s
7. 【STRING 限制】STRING 类型仅支持 ASCII，最大 32767 字符（视 PLC 内存）
8. 【保留区】R9000-R951F 和 DT90000-DT90999 为系统保留，禁止用户程序写入
9. 【END_PROGRAM】每个 PROGRAM/FB 必须以对应的 END_xxx 结尾

## 代码风格要求
- 使用 IF-THEN-ELSIF-ELSE-END_IF 表达主要控制逻辑
- 每个 I/O 地址在 VAR 块声明为 BOOL，后接 // 中文注释
- 急停逻辑必须放在 PROGRAM 的最前面，独立 IF 块
```

```
- 所有输出必须有明确的 FALSE 复位条件
- 自锁电路必须包含解锁条件

## 示例（必须遵循此格式）
PROGRAM Demo
VAR
  X0 : BOOL; // 启动按钮
  X1 : BOOL; // 停止按钮
  X2 : BOOL; // 急停开关（常闭）
  Y0 : BOOL; // 电机运行
  TON_Delay : TON; // 延时定时器实例
END_VAR

// 急停优先
IF NOT X2 THEN
  Y0 := FALSE;
END_IF;

// 启停控制
IF X0 AND X2 THEN
  Y0 := TRUE;
END_IF;
IF NOT X1 OR NOT X2 THEN
  Y0 := FALSE;
END_IF;

// 延时关断
TON_Delay(IN := NOT Y0, PT := T#2s);

END_PROGRAM
```

6.2 Improved User Prompt (Complex System)

```
请为 Panasonic FP-XH PLC 编写一个【多工位自动装配线】控制程序。

## I/O 清单（以下地址必须在 VAR 块声明并在代码逻辑中使用，不可替换别名）
### 输入 (X)
- X0: 急停按钮 (常闭)
- X1: 启动按钮 (常开)
- X2: 停止按钮 (常开)
- X3: 复位按钮 (常开)
- X4: 上料工位物料检测光电传感器
- X5: 上料工位气缸前限位
n...（全部 56 个地址）

### 输出 (Y)
```

- Y0: 主电机运行（传送带）
- Y1: 上料气缸伸出
n...（全部 15 个地址）

内部继电器（R）
- R0: 系统运行中标志
- R1: 上料完成标志
n...（全部 10 个地址）

定时器实例（需要在 VAR 中声明为 TON/TOF/TP）
- TON_*Load* : *TON; // 上料延时 2秒*
- *TON*_Press : TON; // 压装保压 3秒
n...（全部 6 个定时器）

计数器实例（需要在 VAR 中声明为 CTU/CTD）
- CTU_*Total* : *CTU; // 总产量计数*
n...（全部 4 个计数器）

动作流程（按顺序实现）
1. 启动条件检查 → 2. 上料 → 3. 压装 → 4. 检测 → 5. 下料 → 6. 循环/停止
（详细步骤省略，与原始输入一致）

安全条件
- 急停 X0=OFF 时所有输出立即断开
- 安全门 XF/X10 打开时系统暂停
- 气压低 X11 报警时暂停
- *压装超时 TON*_Press 10秒未完成则报警

输出要求
1. 先输出完整的 VAR 块（所有地址和 FB 实例）
2. 再输出急停和安全逻辑（最前面）
3. 再输出主流程逻辑（IF-THEN-ELSIF 按步骤顺序）
4. 最后输出 END_*PROGRAM*
5. 【自检】生成完成后，检查 VAR 中声明的每个 X/Y/R 地址是否都出现在逻辑中。如有遗漏，补充逻辑。
*6. 【自检】确保所有 TON*_/CTU_ 实例都被调用过。

6.3 Fix Prompt (For Validation Feedback Loop)

以下是你之前生成的 Panasonic ST 代码的验证结果。请修复所有问题，输出完整的修正后代码：

发现的问题
- 【结构错误】缺少 END_*PROGRAM 结尾标记*
- *【I/O 未使用】以下声明的地址在逻辑中未使用：X10, X11, X12, X13,* *TON*_Load, CTU_*Total*
- *【安全警告】未检测到互锁逻辑（上料气缸 Y1/Y2 需互锁）*
- *【语法警告】第 45 行赋值语句缺少分号*

```
### 修复要求
1. 不要改变已有的正确逻辑
2. 补充缺失的 END_PROGRAM
3. 为 X10, X11, X12, X13 添加安全门/气压/油温/变频器故障处理逻辑
4. 为 TON_Load 添加上料延时调用：TON_Load(IN := ..., PT := T#2s)
5. 为 CTU_Total 添加产量计数调用：CTU_Total(CU := ..., RESET := ..., PV := 9999)
6. 为 Y1/Y2 添加互锁：Y1 和 Y2 不能同时为 TRUE
7. 修正第 45 行的语法错误

请直接输出修正后的完整代码，不要解释修改内容。
```

---

7. Conclusion

The current PLC-AIStudio AI generation pipeline is **fundamentally constrained by three factors** that together explain the 45/100 score on complex systems:

1. **Hardware limitation**: `moonshot-v1-8k` does not provide sufficient output tokens for industrial-scale ST programs (>50 I/O, >10 steps). This is the **primary cause of truncation**.
2. **Prompt vagueness**: The system prompt lacks Panasonic-specific constraints (no EN/ENO, DDT prefix, instance declarations) and contains **no few-shot examples**, causing the model to hallucinate generic IEC patterns that may not compile in FPWIN Pro7.
3. **Architectural gap**: There is **no program decomposition strategy** and **no validation feedback loop**. The model generates monolithic code without self-checking, and the system does not ask it to fix flagged issues.

Immediate Action Items

Priority	Action	Expected Impact
P0	Switch default model to `moonshot-v1-32k` with `max_completion_tokens=12000`	Eliminates truncation for systems up to ~100 I/O
P0	Rewrite system prompt with Panasonic constraints + few-shot example	Reduces syntax/compilation errors by 50%+
P0	Add POU decomposition for >30 I/O systems	Improves maintainability and reduces per-generation token load
P1	Append I/O coverage self-check instruction to user prompt	Reduces unused-I/O warnings from ~67% to <15%
P1	Implement server-side validation → AI fix loop	Enables automatic recovery from truncation and logic gaps
P2	Add FB templates (MotorLatch, CylinderCtrl, SafetyMgr)	Improves code reusability and safety consistency

Model Roadmap

Phase	Model	Context	Use Case
Immediate	`moonshot-v1-32k`	32K	All industrial generation until volume justifies upgrade
Short-term	`kimi-k2.5`	256K	Very complex systems (>100 I/O, multi-axis positioning)
Long-term	Fine-tuned `kimi-k2-turbo`	128K	Domain-optimized for Panasonic ST after collecting 10K+ validated examples

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Appendices

A. References

1. Panasonic Industry, *Control FPWIN Pro7 Programming Manual* (MN_63489_0050_EN)
2. Panasonic Industry, *Control FPWIN Pro7 Reference Manual* (ACGM0142V50END)
3. Panasonic Industry, *FP0H Series Catalog* (fp0h_e_cata)
4. Panasonic Industry, *FP-XH Series Hardware Manual* (mn_fpxh_hardware_eu_en)
5. Kimi API Platform, *FAQ — Output Length and Truncation* (platform.kimi.ai/docs/guide/faq)
6. IEC 61131-3:2013, *Programmable controllers — Part 3: Programming languages*

B. Glossary

Term	Meaning
POU	Program Organization Unit (PROGRAM, FUNCTION, FUNCTION_BLOCK)
FB	Function Block — reusable logic block with instance memory
EN/ENO	Enable In / Enable Out — conditional execution pins on IEC FBs
DDT	Double Data Register — Panasonic 32-bit register prefix
VAR	Variable declaration block in IEC 61131-3
T/C	Timer / Counter relay area in Panasonic FP series
ST	Structured Text — textual PLC programming language